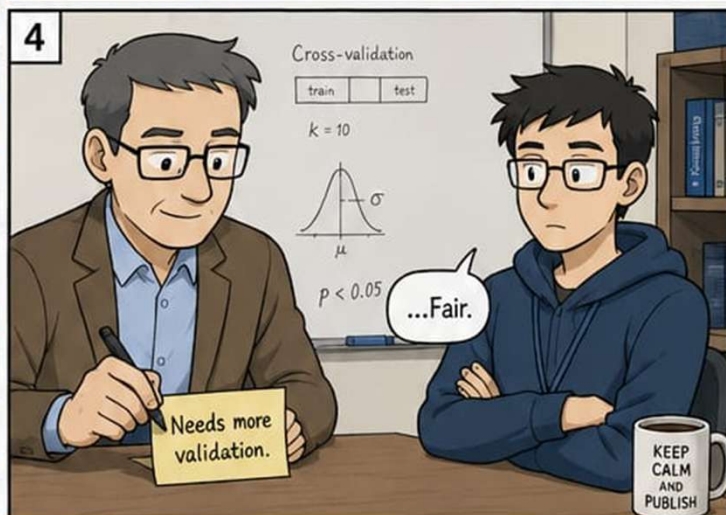
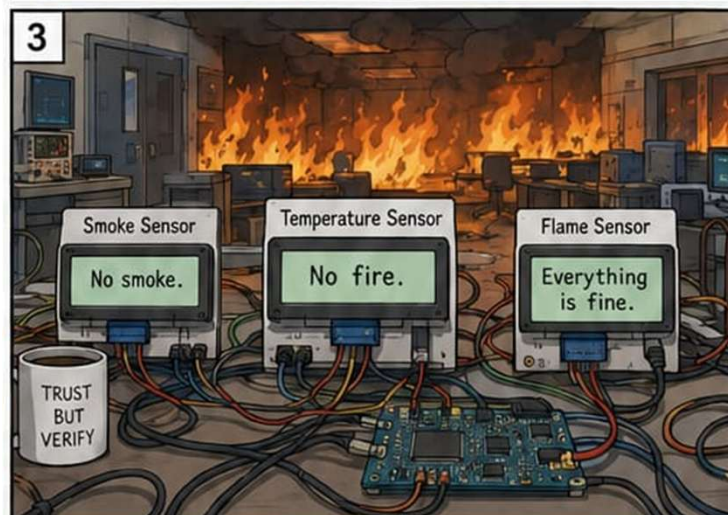
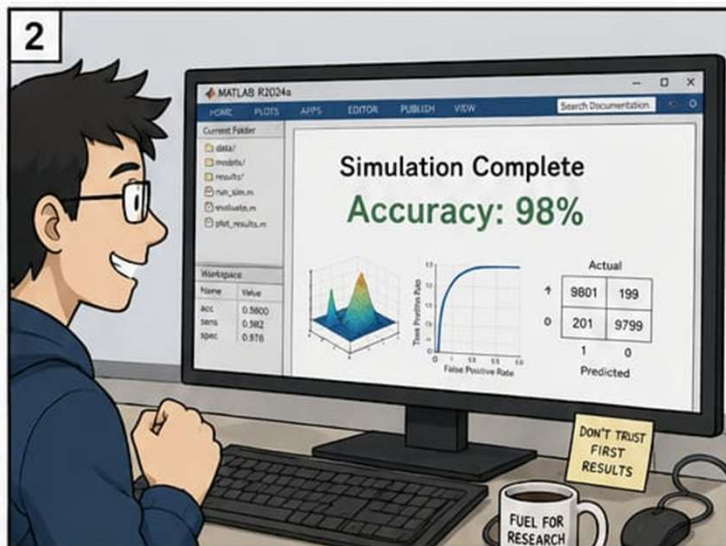
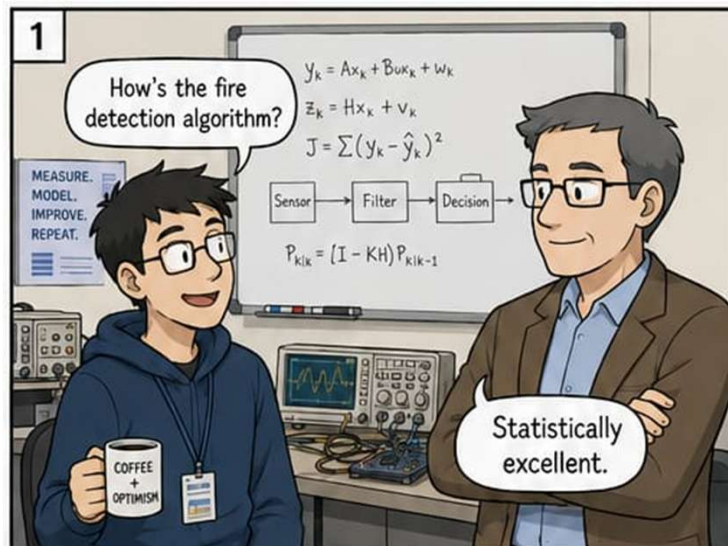






Engineering is the art of solving yesterday's problem just in time for tomorrow's experiment.

Smart Multi-Sensor Fire Detection

One-line signal

This project shows that a simple, well-tuned multi-sensor alarm can become much sharper and more trustworthy when smoke, temperature, and flame evidence are allowed to confirm each other instead of acting alone.

Why this feels promising

There is a strong practical idea underneath the simulation: do not ask one sensor to solve the entire fire problem. Smoke can be noisy, temperature can be slow, and flame can be situational. Together, they create a much fairer decision.

The core build

MATLAB was used to simulate scenarios and optimize thresholds.

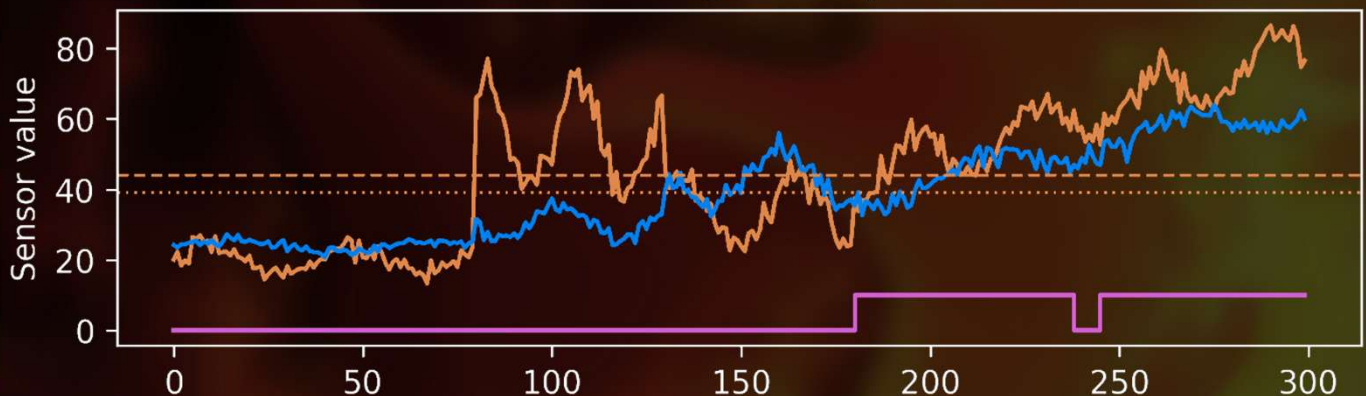
Simulink was used to implement the final decision logic.

Inputs: smoke level, temperature, and flame presence.

Fire rule: smoke and temperature both cross threshold, or flame appears together with elevated smoke.

Alarm latch: once fire is detected, the alarm stays on until reset.

Simulated sensor readings and optimized thresholds



What stands out

Single-sensor logic was noticeably weaker.

A simple OR rule was fast but far too alarm-happy.

Threshold optimization made the multi-sensor logic more defensible.

The latch solved an important safety behaviour issue by preventing brief alarm dropouts during a real fire sequence.

Measure	Final optimized + latched model
Accuracy	98.0%
False alarm rate	0.0%
Missed detection rate	5.0%
Response time	6 s

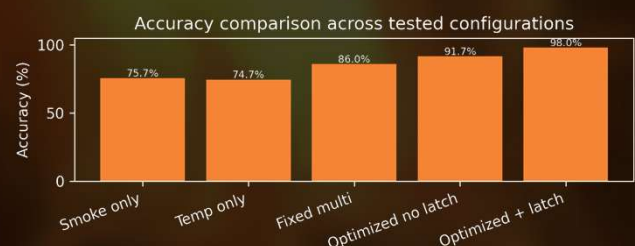
This is the kind of project that could grow well. With real sensors, controlled lab trials, and nuisance-source testing, it has a clear path from a careful simulation exercise to a credible embedded fire-detection prototype.

Where the paper is strongest

The strongest claim is not that this is a finished commercial detector. It is that transparent logic, tuned carefully, can close part of the gap between crude fixed-threshold systems and heavier AI classifiers.

Honest boundary

The signals were simulated, not measured in a live fire environment. So the result should be read as a strong algorithm-validation study, with real hope for hardware testing later, not as proof of field-ready detector performance.

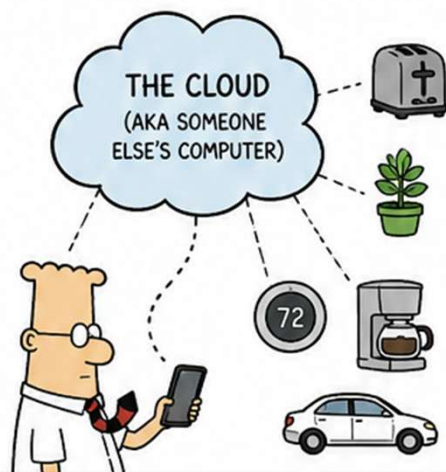


Deone Zarate

THE INTERNET OF THINGS (IoT)

CONNECTING EVERYTHING.
UNDERSTANDING VERY LITTLE.

A DILBERT GUIDE TO IoT REALITY



IOT PROMISE:

- ☒ SMARTER DEVICES
- ☒ REAL-TIME DATA
- ☒ BETTER DECISIONS
- ☒ A BRIGHTER FUTURE

IOT REALITY:

- ☒ MORE ALERTS
- ☒ MORE DASHBOARDS
- ☒ MORE PASSWORDS
- ☒ A BRIGHTER FIREWALL

WITH IoT, WE CAN
CONNECT ANYTHING
TO THE INTERNET!

EVEN IF IT
SHOULDN'T
BE.

STEP 1: CONNECT IT
WE PUT WI-FI IN OUR TOASTER
SO IT CAN TELL US OUR BREAD
IS DONE.

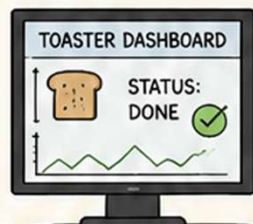
IT'S A
TOASTER.
NOT A
PHILOSOPHER.

STEP 2: SEND IT TO THE CLOUD
THEN IT SENDS DATA TO
"THE CLOUD."

101011
010110
101001

BECAUSE
MY TOASTER
TRUSTS THE
INTERNET.

STEP 3: MAKE A DASHBOARD
WE VISUALIZE THE DATA IN
REAL TIME.



NOTHING
SAYS VALUE
LIKE A PIE
CHART.

STEP 4: GET INSIGHTS
OUR AI ANALYZE THE DATA AND
PROVIDE ACTIONABLE INSIGHTS.

INSIGHT:
YOU EAT
TOO MUCH
BREAD.

STEP 5: CHANGE THE WORLD
WE USE THE INSIGHTS TO DRIVE
DIGITAL TRANSFORMATION.

OR WE
ADD ANOTHER
DASHBOARD.
BOTH ARE
POSSIBLE.

IOT SECURITY

EVERY DEVICE IS A POTENTIAL
DOORWAY.

I HACKED
YOUR PLANT.
IT TOLD ME
ITS WIFI
PASSWORD.

I JUST
WANTED
ATTENTION.

IOT PROJECT MEETING

WE HAVE BIG PLANS AND A SMALL
BUDGET.

WE WANT GOOGLE SCALE
RESULTS WITH A \$500
SENSOR AND A DREAM.

MAINTENANCE

ONCE DEPLOYED, NO ONE
OWNS IT.

OUR IOT PLATFORM
IS DOING GREAT.

NO ONE
TOUCHES IT.

IOT IN THE REAL WORLD
IT WORKS... UNTIL IT DOESN'T.

FIRMWARE
UPDATE...

ERROR 418:
I'M A TEAPOT.

SO, WHAT IS IoT REALLY?

IT'S NOT JUST TECHNOLOGY.
IT'S PEOPLE, PROCESS, DATA,
AND A SENSE OF HUMOR.

AND
LOWERED
EXPECTATIONS.

FINAL THOUGHT

IoT ISN'T ABOUT CONNECTING
THINGS.

IT'S ABOUT CONNECTING
IDEAS... AND PRAYING
THE WI-FI HOLDS.

BLESS THE
SIGNAL
STRENGTH.

REMEMBER:
IN IoT, DATA IS KING.
BUT CONTEXT IS THE
QUEEN WHO MAKES
HIM USEFUL.



NEXT PAGE:

AI AND IoT EV BATTERY
MONITORING



SENSORS ARE CHEAP.
BANDWIDTH IS EXPENSIVE.
BUT MAKING SENSE OF IT?
PRICELESS.



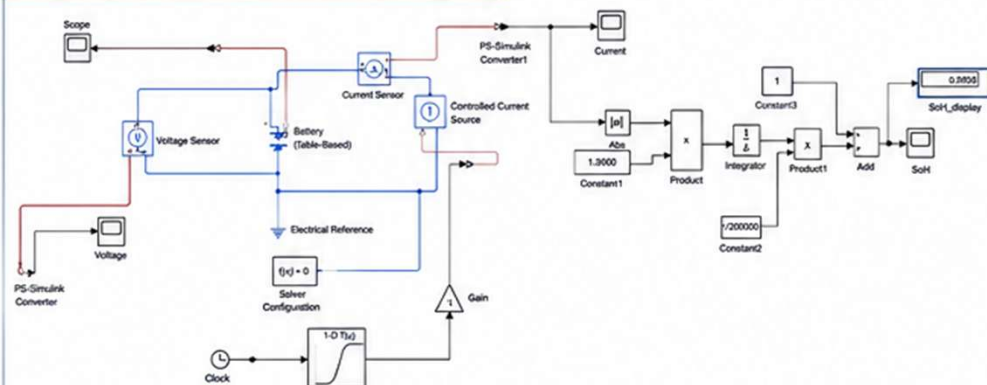
AI AND IOT EV BATTERY MONITORING

From Sensing to Cloud to
State-of-Health Prediction



“A clean proof-of-concept that makes battery data visible, connected, and analytically useful.”

ONE-LINE SIGNAL ARCHITECTURE



Simulink architecture of the ESP32-based EV battery monitoring system from sensing to SOH estimation.



THE CORE WORKFLOW

1. Wokwi was used to simulate an ESP32-based sensing setup.
2. Voltage, current, and temperature were uploaded through Wi-Fi to ThingSpeak.
3. SOC was derived from the monitored signals.
4. MATLAB and Simulink were used for SOH-related processing and prediction.
5. The study compared healthy, medium, poor, and critical battery-condition scenarios.

WHY THIS FEELS PROMISING



The project does not overcomplicate the architecture. It uses an ESP32, ThingSpeak, and MATLAB in a sequence that is easy to understand, easy to extend, and genuinely relevant to future battery-supervision ideas.

KEY MESSAGE IN PLAIN LANGUAGE



The paper is at its best when read as an integration success. It shows that battery telemetry, cloud dashboards, and a prediction stage can all live in one coherent workflow without expensive hardware.

WHAT TO CELEBRATE CAREFULLY



The hopeful part is not that battery prognostics has been solved. The hopeful part is that the pipeline works end to end. That matters because many student projects stop at sensing or stop at dashboards. This one pushes through to analytics.

RESULTS WORTH KEEPING

- ✓ The dashboard clearly separated healthier and degraded operating states.
- ✓ Rising temperature and falling SOC created an intuitive warning pattern.
- ✓ A regression-based SOH prediction stage ran successfully on exported data.
- ✓ Reported RMSE was low, but the prediction range and dataset were both very small.



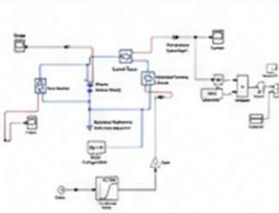
HONEST BOUNDARY



The inputs were simulated, and the SOH labels came from a simplified cycle-count rule rather than experimental battery aging data. That means the prediction result is best treated as feasibility evidence, not validated prognostics performance.

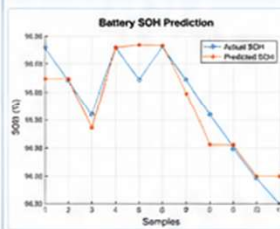
VISUAL SPREAD – FROM DATA TO INSIGHT

1. ESP32 SIMULATION ARCHITECTURE



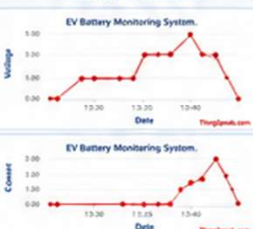
The emulated hardware layer that feeds the rest of the workflow.

2. MATLAB & SIMULINK SOH MODEL



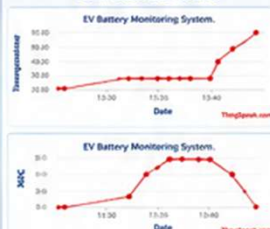
The analytics stage that turns telemetry into health-related interpretation.

3. THINGSPEAK DASHBOARD VOLTAGE & CURRENT



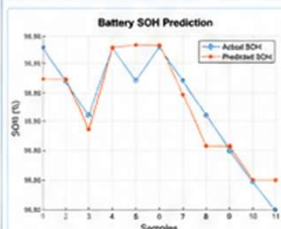
Cloud plots make condition changes visible rather than hidden in raw data streams.

4. THINGSPEAK DASHBOARD TEMPERATURE & SOC



Temperature rise and SOC decline provide the most readable operating story.

5. ACTUAL vs PREDICTED SOH



A useful first prediction view, even if the dataset is still too small for strong claims.

BULLETIN TAKEAWAY

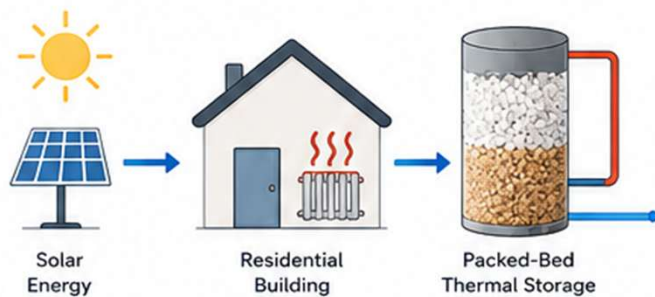
This project opens a practical door. With real battery data, stronger labels, and larger operating histories, the same architecture could evolve from a smart student prototype into a serious small-scale battery-health research platform.



BRINE-DERIVED SALT FOR PACKED-BED THERMAL ENERGY STORAGE

A Feasibility Study of Brine Waste as an Alternative Thermal Storage Medium

Turning desalination waste into practical residential thermal storage.



ONE-LINE SIGNAL

This study investigates whether a brine-derived chloride-rich salt proxy can replace sand in a residential packed-bed thermal energy storage (TES) system while maintaining indoor thermal comfort.

WHY THIS MATTERS



Waste Recovery

Reuse desalination by-products instead of disposal.



Residential Energy

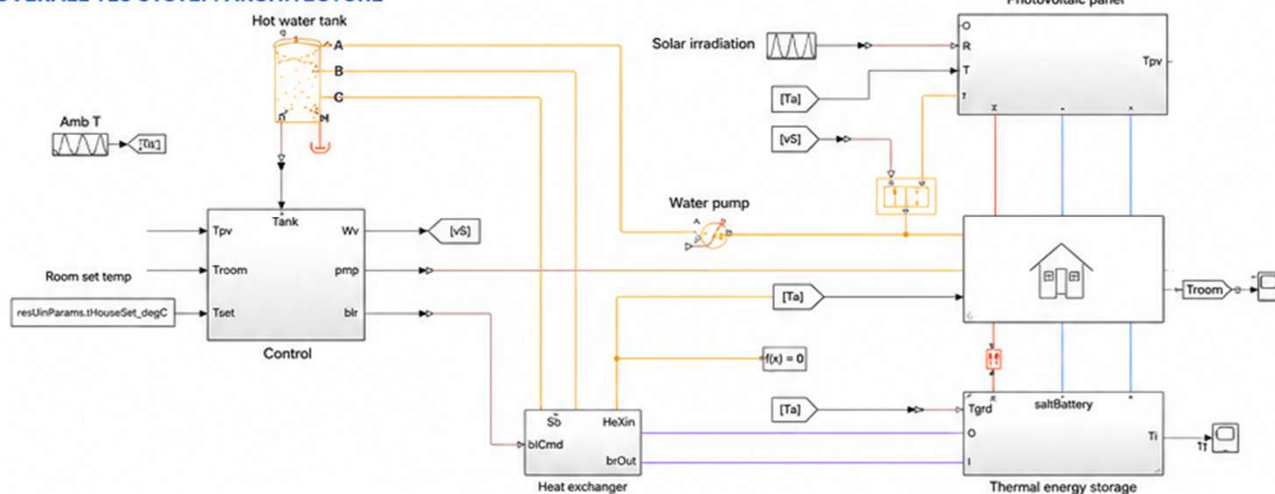
Investigate low-cost seasonal heat storage.



Sustainable Materials

Explore alternative storage media with future circular-economy potential.

OVERALL TES SYSTEM ARCHITECTURE



COMPARED STORAGE MEDIA

	Sand	Salt Proxy
Type	Baseline	Alternative
Description	Well-established granular material	Chloride-rich granular material
Behavior	Higher charging temperature	Lower charging temperature

ASSUMED SALT PROPERTIES

Density	1850 kg/m ³	Thermal Conductivity	1.2 W/m-K	Specific Heat	1500 J/kg-K
---------	------------------------	----------------------	-----------	---------------	-------------

SIMULATION CONDITIONS



- ✓ Same residential TES model
- ✓ Same geometry
- ✓ Same controls
- ✓ Same 24-hour simulation

Only storage medium properties changed.

KEY FINDINGS



Sand storage reached approximately **74°C**.



Salt proxy stabilized near **49°C**.

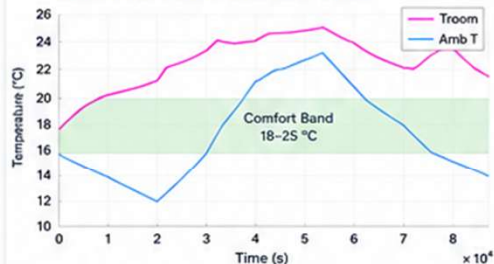


Indoor temperature remained within the **18–25°C** comfort band.



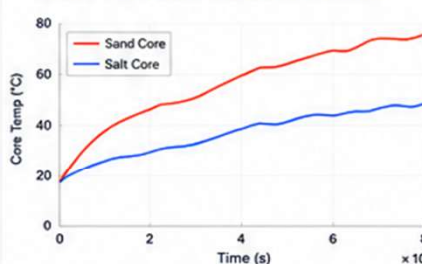
The replacement changed storage behavior but did not significantly affect room comfort.

ROOM TEMPERATURE RESPONSE



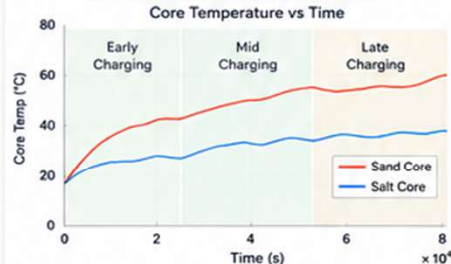
Indoor temperature stayed within the desired comfort range despite lower storage temperature.

CORE TEMPERATURE COMPARISON



Sand storage reached a higher core temperature, while salt proxy charged more gently.

CHARGING CHARACTERISTICS



Different charging behavior: sand heats faster; salt proxy shows a slower, steadier rise.

ENGINEERING INSIGHT

“Higher storage temperature does not necessarily translate into better residential performance. Controlled, comfort-preserving heat delivery is often more valuable than maximizing storage temperature.”

STUDY BOUNDARY

This work is an early-stage simulation using a proxy material.

Not included:

- Experimental material characterization
- Corrosion analysis
- Moisture behaviour
- Cycling durability
- Brine recovery energy

Therefore, the results demonstrate feasibility, not deployment readiness.



BULLETIN TAKEAWAY

This study provides encouraging evidence that brine-derived salt deserves further laboratory investigation as a low-cost thermal storage medium. Future work should validate material properties experimentally and evaluate long-term cycling performance.

OPTIMIZING SOLAR HARVEST

Dust-Aware PV Tilt Optimization for Rooftop Solar in Abu Dhabi



ONE-LINE SIGNAL

This study demonstrates that carefully selecting the tilt angle of rooftop photovoltaic panels—while accounting for desert dust and high operating temperatures—can substantially improve annual energy production, financial returns, and carbon savings for commercial buildings in Abu Dhabi.



WHY THIS FEELS PROMISING

Instead of assuming a standard tilt, this work uses PVSyst simulations with local weather and soiling conditions to determine the best orientation specifically for Abu Dhabi.

The result is a methodology that designers can reuse for similar commercial rooftop installations across the UAE.

ENGINEERING CHALLENGE



Extreme Solar Resource
Abu Dhabi has excellent solar irradiation but also very high module temperatures.



Dust & Soiling
Dust accumulation reduces PV output unless the system is cleaned or installed at favourable tilt angles.



Commercial Retrofit
Older glazed office buildings have high cooling loads, making rooftop PV an attractive retrofit option.

Case Study:
ASCORP Holding Office, Abu Dhabi

CORE WORKFLOW

1. Site Survey



2. PVSyst System Design



DN3-100KTL

3. Tilt & Azimuth Scan



4. 8760-hour Simulation



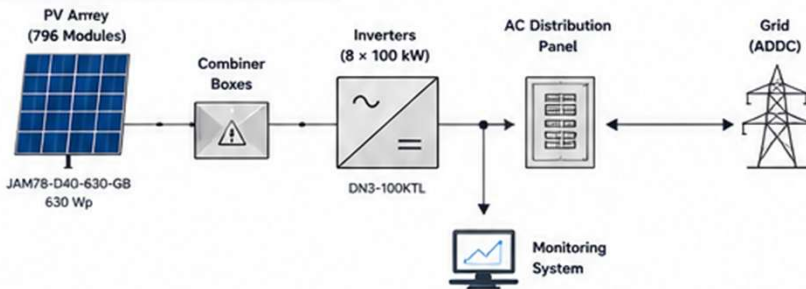
5. Financial Analysis



6. Carbon Assessment



PVSYST SYSTEM ARCHITECTURE



Simulation workflow used to optimize rooftop PV orientation for ASCORP Holding.

SIMULATION SETTINGS

- Location: Abu Dhabi (24.3343°N, 54.4894°E)
- Weather Data: Meteonorm 9.0
- Soiling Losses: 3–5% (quarterly cleaning)
- Temperature Model: NOCT with local wind
- Degradation: 1.0%/year (linear)
- Transposition Model: Perez diffuse
- Simulation: 8760 hourly steps (1 year)

OPTIMAL ORIENTATION



Tilt Angle
26.3°
Azimuth
0° (South)

Global on Collector
2,285 kWh/m²
Transposition Factor
1.08

This orientation delivered the highest annual solar collection.

ENERGY PERFORMANCE



Annual Energy
893 MWh/year

Performance Ratio (PR)

0.783

Capacity Factor

20.4%

Specific Production

1,783 kWh/kW

Performance ratio exceeds regional benchmarks (0.75–0.80).

FINANCIAL PERFORMANCE



LCOE
0.0351 USD/kWh

Payback Period

5.2 years

25-year Net Profit

\$2.25 Million

Approximately 65% cheaper than the commercial grid tariff.

ENVIRONMENTAL IMPACT



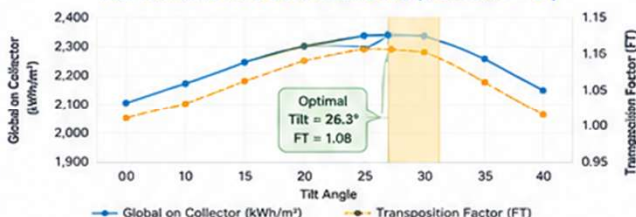
Annual CO₂ Reduction
438.7 tonnes

Lifetime CO₂ Reduction (25 years)

13,161 tonnes

Equivalent to removing ~95 passenger vehicles from the road each year.

TILT PARAMETRIC SCAN RESULTS (AZIMUTH = 0°)



26.3° tilt angle provides the highest annual collection (2,285 kWh/m²), achieving an 8.1% gain compared to horizontal mounting.

COMPARISON WITH LITERATURE BENCHMARKS

Study	System Type	Specific Yield (kWh/kW)	Performance Ratio (PR)	Payback (Years)
This Study	Rooftop (501 kW)	1,783	0.783	5.2
Hasanain (2024) [1]	Ground-mounted	1,800 – 1,950	–	–
Elnabawi (2021) [4]	ESTIDAMA Pearl 1	–	0.75 – 0.80	–
Masdar N21 (2023) [7]	New Construction	1,750	0.78	–

Performance is competitive with literature values despite accounting for temperature derating and 3–5% soiling losses.

KEY TAKEAWAYS

- ✓ Dust-aware optimization identified 26.3° as the optimal tilt angle.
- ✓ 8% increase in energy yield compared to horizontal installation.
- ✓ Strong economics: 5.2-year payback and 65% lower LCOE than grid.
- ✓ High performance ratio (0.783) above regional benchmarks.
- ✓ Significant CO₂ reduction supporting UAE Net Zero 2050 target.

STUDY BOUNDARY

- Simulation-based design (not yet operational).
- Single-year weather data (Meteonorm).
- Simplified shading analysis.
- Zero discount rate assumed.
- Quarterly cleaning schedule assumed.
- No field validation yet.

Results demonstrate potential and feasibility, not operational performance.

BULLETIN TAKEAWAY

This study shows that location-specific PV design matters. By incorporating local dust behavior, high temperatures, and real-world conditions into tilt and azimuth selection, commercial rooftop systems can deliver higher energy, stronger financial returns, and meaningful environmental benefits.

ACKNOWLEDGMENT

The authors thank Ms. Anu Lonappan and Dr. Prasanthi Achikkulaath for their supervision and support, and the University of Greater Manchester Ras Al Khaimah Academic Centre for providing the academic resources required for this project.

HARMONIC ANALYSIS OF MULTILEVEL INVERTERS

Comparing 3-Level, 5-Level and 7-Level Inverters Using SPWM Technique



One-line signal:

Higher-level multilevel inverters with SPWM produce smoother output waveforms and significantly reduce harmonic distortion for better power quality.

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WHY THIS MATTERS



Multilevel inverters generate stepped voltages closer to a sinusoid, reducing harmonics and switching stress.



Lower THD improves power quality, efficiency, and reliability of power electronic systems.



Essential for industrial drives, renewables, UPS, and high-power applications.

KEY TAKEAWAYS

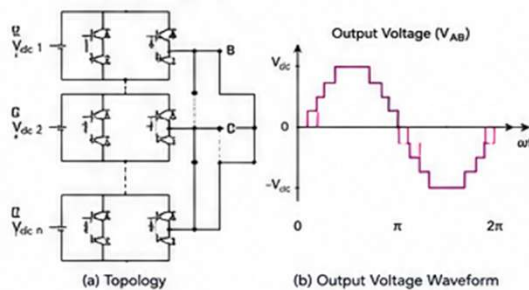
- As the number of levels increases, THD reduces significantly.
- 7-level inverter with Level-Shifted SPWM delivers the lowest THD (2.55%).
- Multilevel inverters enhance efficiency and power quality compared to conventional inverters.

CORE WORKFLOW



CHB MULTILEVEL INVERTER TOPOLOGY

Cascaded H-Bridge Inverter

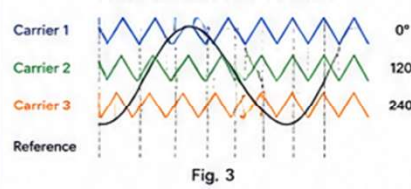


Output Levels for Different Configurations

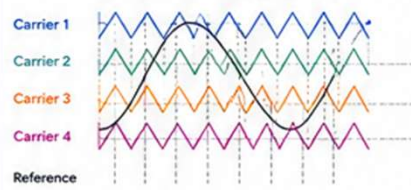
Inverter Level	No. of H-Bridge Cells	DC Source per Cell	Output Voltage Levels
3-Level	1	V_{dc}	$-V_{dc}, 0, +V_{dc}$
5-Level	2	$V_{dc}/2$	$-V_{dc}, -V_{dc}/2, 0, +V_{dc}/2, +V_{dc}$
7-Level	3	$V_{dc}/3$	$-V_{dc}, -2V_{dc}/3, -V_{dc}/3, 0, V_{dc}/3, 2V_{dc}/3, +V_{dc}$

SPWM TECHNIQUES

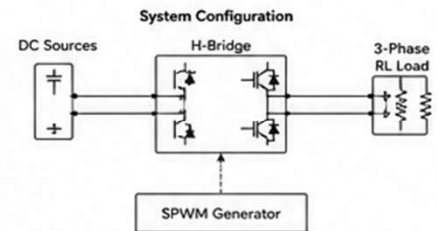
Phase-Shifted PWM (PS-PWM)



Level-Shifted PWM (LS-PWM)



SIMULATION SETUP

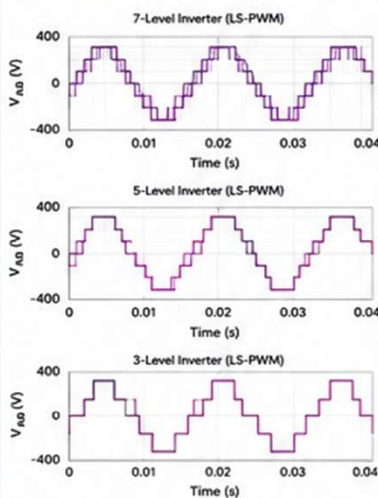


Key Components

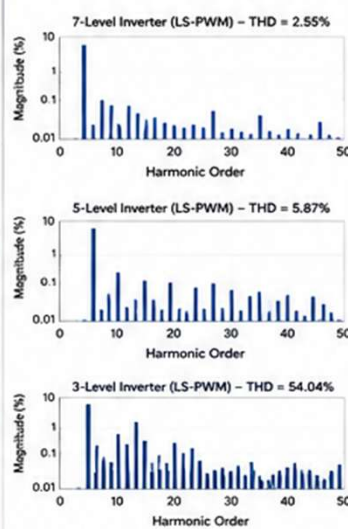
- 3-Level, 5-Level, 7-Level CHB Inverters
- Modulation: LS-PWM & PS-PWM
- Load: 3-Phase R-L Load (Same for all)
- Analysis Tool: FFT (Powergui)
- Software: MATLAB/Simulink

SIMULATION RESULTS & FFT ANALYSIS

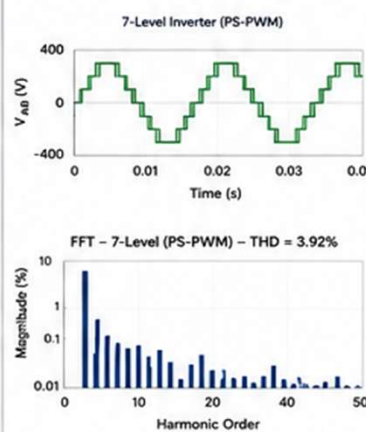
LINE-TO-LINE VOLTAGE WAVEFORMS (LS-PWM)



FFT ANALYSIS (LS-PWM)



FFT ANALYSIS (PS-PWM, 7-LEVEL)



COMPARISON SUMMARY

Inverter Level	Modulation Technique	Fundamental Magnitude (%)	THD (%)
3-Level	LS-PWM	314.4	54.04
	PS-PWM	316.0	53.72
5-Level	LS-PWM	228.2	5.87
	PS-PWM	273.0	5.43
7-Level	LS-PWM	468.4	2.55
	PS-PWM	548.1	3.92

Waveform Quality Assessment

3-Level	Poor	High harmonic distortion
5-Level	Moderate to Good	Reduced harmonics
7-Level	Very Good	Near-sinusoidal output

Waveform becomes smoother as the number of levels increases, reducing harmonic distortion.

LS-PWM outperforms PS-PWM for 7-level inverter with the lowest THD.

CONCLUSION

- Increasing the number of voltage levels in CHB multilevel inverters significantly reduces THD.
- Among all configurations, the 7-level inverter with LS-PWM achieved the lowest THD of 2.55%.
- Higher-level multilevel inverters provide superior power quality, improved efficiency, and lower harmonic pollution.
- These characteristics make them ideal for industrial drives, renewable energy systems, UPS, and other high-power applications.

APPLICATIONS

- Industrial Motor Drives
- Renewable Energy Systems
- HVAC & UPS Systems
- FACTS & Power Conditioning
- Grid-Connected Systems

FUTURE WORK

- Experimental validation of hardware prototypes.
- Investigation of other modulation methods such as SHE-PWM and SVPWM.
- Analysis under varying load and unbalanced conditions.
- Efficiency and switching loss comparison.

REFERENCES

For detailed reference list, please refer to the full paper. Key references include works from IEEE Transactions, IET Power Electronics, and International Journals.



BULLETIN TAKEAWAY

More levels. Less distortion. Better power quality.
7-level CHB inverter with LS-PWM is the best performer.

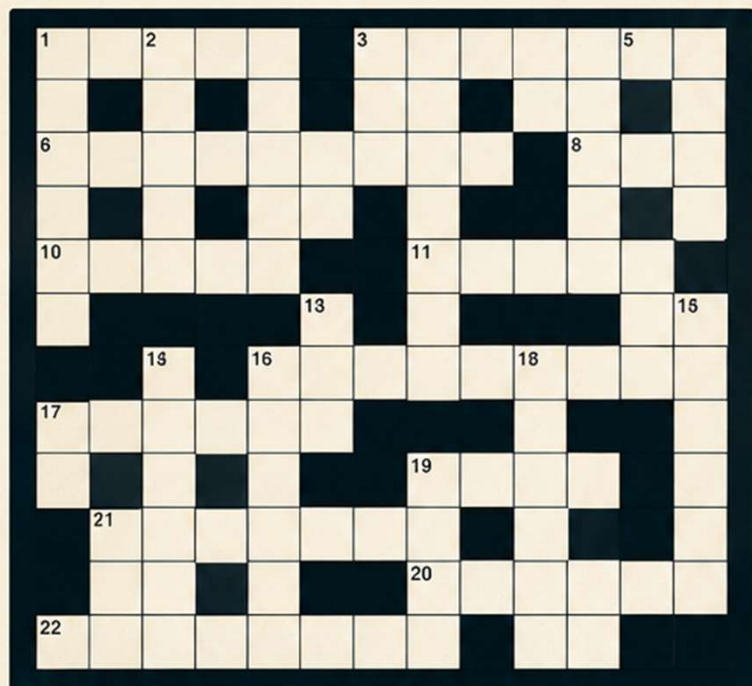




CROSSWORD

TEST YOUR KNOWLEDGE

Clues inspired by technology, sustainability, and the research themes in this bulletin.



ACROSS

- 1 Abbreviation for Internet of Things
- 6 Cloud platform used for IoT dashboards
- 8 Unit of electric current
- 9 Metric for battery health
- 11 ESP32 is a popular _____
- 13 Symbol for degrees Celsius
- 15 Sand is the baseline in the TES _____
- 16 The main source of solar energy
- 17 Tilt angle that maximizes PV yield
- 19 Software used for PV system simulation
- 20 Gas produced during combustion
- 21 District in Abu Dhabi where this case study is set
- 22 Measure of energy produced per unit area

DOWN

- 1 State of Charge (abbr.)
- 2 A type of renewable energy
- 3 Opposite of discharge
- 4 Material proposed as salt proxy in TES
- 5 Losses due to dust on PV panels
- 7 Temperature of a room
- 10 Residence or building
- 12 Storage system for thermal energy
- 14 Carbon dioxide (chemical formula)
- 16 Simulation tool used for circuit modeling
- 18 Direct current (abbr.)

LAST ISSUE ANSWERS



ACROSS:
1 IOT, 6 THINGSPEAK, 7 VOLTAGE, 8 CURRENT, 9 SOC, 10 MATLAB, 11 SOH, 12 SIMULINK, 14 PREDICT, 15 TEMPERATURE, 17 BATTERY, 18 ESP32, 19 REGRESSION, 20 WIFI, 21 CLOUD, 23 PIPELINE, 24 DASHBOARD, 25 ANALYTICS, 26 TELEMETRY, 28 CONNECT, 29 CHARGE, 30 AH, 31 SAND, 32 THINGSPEAK

DOWN:
1 IV, 2 OVERCURRENT, 3 GSM, 4 TELEMETRY, 5 SCOPE, 6 TCP, 13 MODELLING, 16 REFERENCE, 18 EXPORT, 20 WIRELESS, 22 DEVICE, 27 ALERT

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CHOOSE EXOTIC. CHOOSE BERGAMONT.



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1 tsp (2g) per cup • 90°C water • Steep 3–5 min

Sip slowly. Live fully.



Learn. Explore. Innovate.
See you in the next edition!

DESIGN AND SIMULATION OF A SOLAR-POWERED RESIDENTIAL EV CHARGING SYSTEM

WITH BATTERY ENERGY STORAGE AND VSS-INC MPPT CONTROL

A Smart, Grid-Friendly Solution for Villas in Ras Al Khaimah, UAE



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ONE-LINE SIGNAL



This study presents a solar-powered smart EV charging system for residential villas with battery storage and grid support. A variable step-size incremental conductance (VSS-INC) MPPT algorithm is compared with conventional INC under identical conditions.

WHY IT MATTERS

- ✓ Reduces dependency on the utility grid for EV charging.
- ✓ Maximizes rooftop solar self-consumption.
- ✓ Battery storage ensures reliable charging and load support.
- ✓ Supports UAE's clean energy and sustainable transport goals.

KEY HIGHLIGHTS



Solar Priority
PV is the primary energy source.



Smart Storage
Excess solar stored, used when needed.

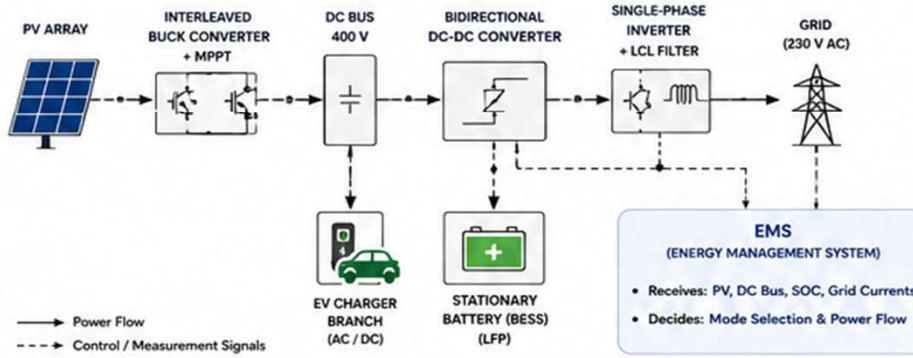


Grid Support
Grid used only when required.



Better Tracking
VSS-INC captures more solar power with lower ripple.

SYSTEM ARCHITECTURE



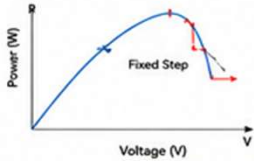
4-MODE ENERGY MANAGEMENT SYSTEM (EMS)

1	MODE 1 Solar EV Charging	- EV connected and PV available - PV powers EV + DC load; grid only at very low irradiance
2	MODE 2 Grid-Assisted EV Charging	- EV connected, PV unavailable - Grid supplies EV via inverter
3	MODE 3 Solar Battery Charging	- EV not connected, PV available - PV charges BESS and supplies DC load
4	MODE 4 BESS Discharge	- EV not connected, PV unavailable - BESS discharges to support DC load

MPPT ALGORITHMS COMPARED

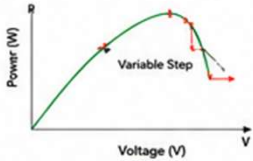
INC (Incremental Conductance)

Fixed step-size perturbation. Simple and robust, but slower response and higher steady-state ripple.



VSS-INC (Variable Step-Size INC)

Adaptive step-size based on distance from MPP. Faster tracking with lower ripple near MPP.



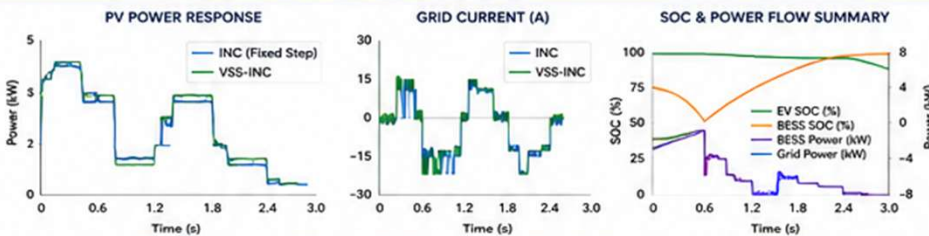
BENEFITS OF VSS-INC

- ⚡ Faster convergence to MPP under irradiance changes
- 📉 Lower steady-state ripple around MPP
- ⚙️ Better energy extraction from rooftop PV

SYSTEM & SIMULATION SETUP

DC Bus Voltage	400 V (nominal)
PV Irradiance Profile	1000 – 500 – 10 – 500 – 1000 W/m ²
MPPT Cases	INC and VSS-INC
Simulation Time	3 seconds
Battery Type (BESS)	LFP
Main Outputs	PV P-V-I, Grid Current, SOC, Power Flows
Software	MATLAB / Simulink

KEY RESULTS: INC vs VSS-INC MPPT



★ VSS-INC shows faster response to irradiance changes and lower ripple around the MPP, while maintaining the same EMS behavior and grid support logic.

PERFORMANCE COMPARISON

Criterion	INC (Fixed Step)	VSS-INC (Variable Step)	Observation
MPP Response	Slower	Faster	Better tracking with VSS-INC
Steady-State Ripple	Higher	Lower	Reduced ripple near MPP
EMS Behavior	Correct	Correct	No change in power flow logic
Complexity	Low	Moderate	Acceptable for residential use
Best Use	Stable irradiance	Dynamic rooftop irradiance	VSS-INC preferred for villas

FULL-SCALE CONCEPT (VILLA)

PV Array	~ 9.9 kWp (18 high-power modules)
Battery Storage (BESS)	~ 20 kWh (LFP)
EV Chargers	2 x 7 kW Level-2 AC Chargers
Location	Ras Al Khaimah, UAE
Objective	Minimize grid import, maximize solar self-consumption

DESIGN ADVANTAGES

- ✓ Interleaved buck converter reduces input/output ripple
- ✓ Bidirectional converter enables BESS charge/discharge
- ✓ 400 V DC bus for efficient power flow integration
- ✓ Modular architecture for easy scalability



LIMITATIONS

- ⚠️ Simulation duration is short (3 s)
- ⚠️ Simplified battery aging model
- ⚠️ No full-day load or irradiance data
- ⚠️ Grid harmonic and EMC compliance not evaluated
- ⚠️ Hardware validation required before deployment

FUTURE WORK

- 🔧 Full-day simulation with real irradiance and load profiles
- 🔧 Detailed battery aging and thermal analysis
- 🔧 Hardware-in-the-loop (HIL) testing
- 🔧 PCC harmonic & protection assessment
- 🔧 Explore controlled V2G in future phases



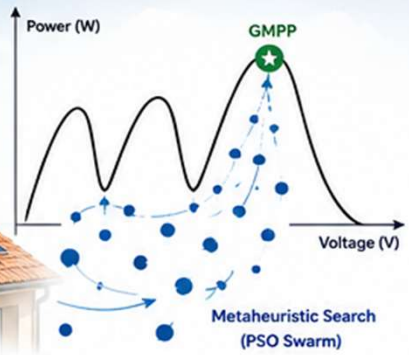
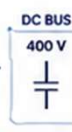
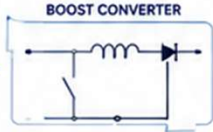
BULLETIN TAKEAWAY

VSS-INC MPPT significantly improves PV energy extraction under dynamic rooftop conditions without altering the EMS or power-flow logic. The proposed architecture is a reliable, scalable, and grid-friendly solution for residential solar EV charging in the UAE.



METAHEURISTIC MPPT FOR PHOTOVOLTAIC SYSTEMS

Improving Global Maximum Power Point Tracking Under Partial Shading Conditions



ONE-LINE SIGNAL

This research evaluates Particle Swarm Optimization (PSO) and Grey Wolf Optimization (GWO) against the conventional Perturb & Observe (P&O) MPPT algorithm to improve Global Maximum Power Point (GMPP) tracking under partial shading conditions using MATLAB/Simulink.

WHY THIS MATTERS



PARTIAL SHADING

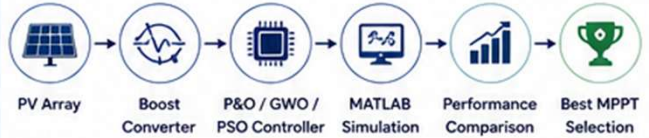
Clouds, dirt and obstacles create multiple power peaks, making conventional MPPT miss the true maximum.



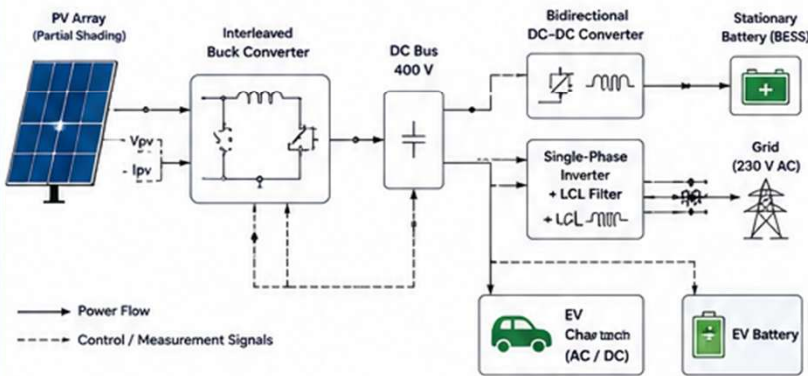
AI-BASED OPTIMIZATION

Metaheuristic algorithms search globally instead of getting trapped in local maxima.

CORE WORKFLOW



MATLAB / SIMULINK SYSTEM ARCHITECTURE



Simulation platform used to compare conventional and metaheuristic MPPT controllers under identical conditions.

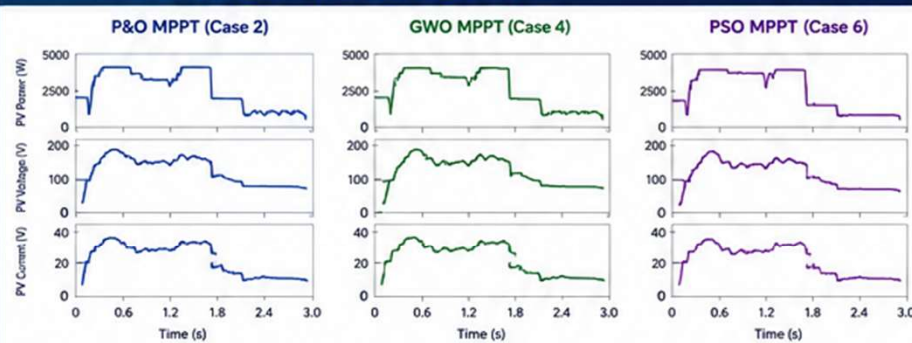
ALGORITHM COMPARISON

P&O (Conventional)	GWO (Grey Wolf Optimization)	PSO (Particle Swarm Optimization)
✓ Simple and easy to implement ✓ Low computational cost ✗ Can get trapped in local maxima under PSC ✗ Higher steady-state oscillations	✓ Better global search capability ✓ Reduced oscillations around MPP ✓ Improved tracking accuracy ✗ Performance depends on parameter tuning	✓ Fastest convergence ✓ Highest power extraction ✓ Stable voltage response ✓ Minimal oscillation ✓ Reliable GMPP tracking
		RECOMMENDED CONTROLLER

PARTIAL SHADING CASES (IRRADIANCE DISTRIBUTION ON TWO PV STRINGS)

Case 1 (P&O)	Case 2 (P&O)	Case 3 (GWO)	Case 4 (GWO)	Case 5 (PSO)	Case 6 (PSO)
1000 W/m ² / 500 W/m ²	1000 W/m ² / 200 W/m ²	1000 W/m ² / 500 W/m ²	1000 W/m ² / 200 W/m ²	1000 W/m ² / 500 W/m ²	1000 W/m ² / 200 W/m ²

KEY SIMULATION RESULTS (EXAMPLES)



SUMMARY COMPARISON (ALL CASES)

Case	Method	Irradiance (W/m ²)	Peak Power (W)	Tracking Accuracy	Stability
1	P&O	1000 / 500	~ 3600	Moderate	●●●●●
2	P&O	1000 / 200	~ 3000	Low	●●●●●
3	GWO	1000 / 500	~ 4200	High	●●●●●
4	GWO	1000 / 200	~ 4100	High	●●●●●
5	PSO	1000 / 500	~ 4350	Very High	●●●●●
6	PSO	1000 / 200	~ 4350	Very High	●●●●●

PSO delivers the highest power output and the most reliable GMPP tracking under severe partial shading.



ENGINEERING INSIGHT

Under partial shading, multiple power peaks appear in the P-V curve. Metaheuristic algorithms explore the entire search space, enabling the controller to find the true Global Maximum Power Point.



LIMITATIONS

- Simulation duration limited to 3 seconds.
- Simplified battery and grid models.
- Real-world testing and long-term validation are required before hardware deployment.



FUTURE WORK

- Hardware-in-the-loop (HIL) implementation.
- Evaluation with real PV arrays and weather data.
- Integration with advanced battery management and smart grid functions.



BULLETIN TAKEAWAY

Metaheuristic MPPT—especially PSO—significantly enhances power extraction from photovoltaic systems under partial shading, ensuring higher energy yield, better stability and reliable operation for next-generation solar energy solutions.





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